AI Network Troubleshooter Report

Filename: advanced_capture.pcap
Packet Count: 3150

Protocols

TCP: 2899
NTP: 12
ARP: 10
DNS: 50
HTTP: 9
TLSv1: 3
TLSv1.3: 87
ICMP: 20
MDNS: 4
UDP: 56

AI Diagnostic Report

1. Root Cause Summary
The packet capture reveals a significant discrepancy between the number of TCP SYN packets (1040) and SYN-ACK packets (10), indicating approximately 1030 possible failed TCP connection attempts. This pattern strongly suggests either a network connectivity issue affecting the TCP handshake completion or a targeted scan/attack such as a SYN flood or reconnaissance activity against the monitored host(s). The top talker, IP 172.17.208.219, is responsible for most traffic, which might be the origin of these connection attempts.

2. Security Observations
- The abnormally high number of TCP connection attempts with very few successful handshakes points to either a network misconfiguration or malicious scanning/attack activity.
- Presence of connections to multiple external IPs such as 45.33.32.156, 104.20.28.246, and 140.82.114.4, including DNS queries to popular domains and suspicious reverse DNS queries (e.g. 156.32.33.45.in-addr.arpa.local), could indicate reconnaissance or external interaction.
- DNS queries to well-known domains like github.com, openai.com, and scanme.nmap.org suggest legitimate or testing traffic; however, the mixture with potential failed TCP connections requires closer inspection.
- TCP SYN flood or connection attempt anomaly is the main concern because it can degrade service availability or be a precursor to a DoS attack.

3. Troubleshooting Recommendations
- Verify network path and firewall/router configuration between the source IP 172.17.208.219 and destination to ensure SYN packets are not dropped or blocked, and SYN-ACKs are returned properly.
- Investigate logs and IDS/IPS alerts for signs of SYN flood attacks or port scans originating from 172.17.208.219 or other IPs.
- Perform endpoint checks on 172.17.208.219 to determine if it is compromised, misconfigured, or running aggressive scanning tools.
- Monitor for unusual spikes in traffic or resource utilization that align with the time of capture.
- Validate DNS and reverse DNS configurations for anomalies and confirm the legitimacy of the requested domains.
- Apply rate limiting or TCP SYN cookies on servers if under attack to mitigate failed connection impact.

- Conduct further packet captures during different time windows for pattern corroboration.

4. Severity Assessment
Severity: High

The high volume of failed TCP connection attempts indicates either a serious network issue impacting connectivity or an early-stage network attack such as SYN flooding or port scanning. This can lead to service disruption or be indicative of reconnaissance before a more damaging attack. Immediate investigation and mitigation steps are required to prevent potential denial-of-service conditions or security breaches.